The One-Electron Universe Hypothesis

From Wheeler & Feynman's 1940 Conjecture to Modern Worldline Topology

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Scope Historical origins, formal QFT treatment, worldline self-intersections, Grassmann variable methods, topological reinterpretations, and open experimental questions. Covers literature from 1941 to 2026.
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Executive Summary

- The hypothesis, first proposed by John Wheeler to Richard Feynman in the spring of **1940**, posits that every electron and positron in the observable universe is a single particle whose worldline weaves forwards and backwards through all of spacetime, intersecting any given constant-time hyperplane an astronomically large number of times.
- The decisive objection Feynman immediately raised is quantitative and still unresolved: the hypothesis demands **equal numbers of electrons and positrons** at every moment of cosmic history, yet the observable universe contains vastly more electrons than positrons — a discrepancy that cannot be argued away without invoking hidden positrons or a mechanism for baryon-number violation.

- The productive legacy of the idea exceeds the idea itself: Feynman "stole" the insight that positrons are electrons traversing worldlines backwards in time, encoding it in his **1949 Theory of Positrons**, which became foundational to modern QED and to the Feynman-diagram representation of pair creation/annihilation.
- In QFT, worldline self-intersections are treated rigorously through the worldline (first-quantised) path integral: closed worldline loops map to one-loop vacuum diagrams, and self-approaching trajectories in Euclidean spacetime are precisely the **worldline instantons** responsible for Schwinger pair production.
- Modern formal revivals using **Grassmann variables** (for spin), knot invariants (Gauss linking number, Jones polynomial), and topological quantum field theory techniques offer a geometrically self-consistent framework that reproduces all standard QED results whilst interpreting fermionic statistics, the Pauli exclusion principle, and vacuum fluctuations as emergent topological consequences of a single braided worldline.

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1. Introduction and Scope

Few ideas in the history of physics illustrate the relationship between bold conjecture and productive formalism more vividly than John Wheeler's **one-electron universe hypothesis**. Proposed in a single telephone call in 1940 and rejected within the same conversation on empirical grounds, the idea nevertheless propagated through the intellectual ecosystem of mid-twentieth-century physics. It seeded one of QED's most powerful conceptual tools — the reinterpretation of antiparticles as particles travelling backwards in time. The worldline perspective that Wheeler's proposal required has since blossomed, through the work of Stueckelberg, Feynman, Nambu, Schwinger, Fock, Bern, Kosower, Strassler, and many others, into a sophisticated first-quantised formalism that computes quantum amplitudes more efficiently than conventional second-quantised Feynman diagram methods in many regimes.

This report traces the full intellectual arc of the one-electron hypothesis: its origin, its immediate formal consequences, the detailed physics of worldline self-intersections, and the modern program of topological and algebraic methods that have turned Wheeler's "tangled knot" metaphor into a rigorous calculational framework. The treatment is aimed at readers with graduate-level familiarity with quantum field theory and differential geometry, though the historical and conceptual sections are accessible to a broader audience. The report covers primary literature from **1941 to 2026**, including recent work on worldline instantons in spatiotemporal fields, the monadic electron framework, and biquaternionic worldline analysis.

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2. The Original Wheeler–Feynman Hypothesis (1940)

2.1 The Telephone Call and the Gedankenexperiment

In the spring of **1940**, John Archibald Wheeler, then a professor at Princeton, telephoned his graduate student Richard Feynman with what he called a fundamental insight. "Feynman," he said, "I know why all electrons have the same charge and the same mass." The answer, he declared, was: "Because they are all the same electron!"

The conceptual starting point is the observation that every electron in the universe is, empirically, absolutely identical to every other — same rest mass of **9.109×10^{-31} kg**, same charge magnitude of **1.602×10^{-19} C**, same spin-1/2. In the particle physics of the late 1930s, this universality was a postulate rather than an explanation. Wheeler proposed that it followed inevitably from the radical claim that there is literally only one electron, and that every electron ever observed is a different segment of its single worldline threading through the entirety of cosmic history. The identical properties are identical because they belong, ultimately, to the same particle.

This is not an outlandish idea within special relativity. A relativistic particle traces a **worldline** — a curve in four-dimensional Minkowski spacetime — as it evolves. The worldline is a geometric object in spacetime; it does not propagate through spacetime the way a ball rolls through a room. Wheeler's proposal was simply that this worldline, instead of having distinct endpoints for each physical electron, was a single connected curve of infinite (or cosmological) length, looped, knotted, and tangled through all of spacetime like an immense thread.

2.2 Worldline Topology and the Time-Slice Argument

Consider taking a "snapshot" of the universe at a fixed cosmic time $t = t_0$. In spacetime language, this snapshot is a spacelike hypersurface $\Sigma(t_0)$. The single worldline, being a curve in four-dimensional spacetime, intersects this hypersurface at multiple points. Each intersection point corresponds to an electron (if the worldline is locally traversed forwards in time) or a positron (if locally traversed backwards in time). The number of forward-moving intersections minus the number of backwards-moving intersections equals a topological invariant of the worldline — the **net winding number** — which Wheeler identified with the conserved total charge of the universe.

If the worldline has n_+ forward-moving intersections and n_- backwards-moving intersections with $\Sigma(t_0)$, then the observable particle content at that moment consists of n_+ electrons and n_- positrons, for a net charge of $-(n_+ - n_-)e$. The simplest consistent scenario is a worldline that begins and ends at the boundary of spacetime (the Big Bang and the ultimate fate of the universe) with no net winding, implying $n_+ = n_-$ — equal numbers of electrons and positrons. This constraint is what immediately doomed the original hypothesis empirically, as Feynman noted at once.

The topology of such a worldline in four dimensions is rich. In 3+1 dimensions, a curve can be knotted and braided, and the intersections with time-slices can be counted using standard tools from differential topology. The **Gauss linking number**, computable for closed curves, and the **Jones polynomial**, a more refined knot invariant, encode geometric information about how the worldline self-entangles. These topological objects will prove central to modern formalisms discussed in Sections 10–11.

2.3 Feynman's Reception and Immediate Objections

Feynman was an ideal interlocutor for this idea — brilliant, mathematically fearless, and constitutionally sceptical of ornate hypotheses not grounded in calculation. His response was swift and lethal: the hypothesis required equal numbers of electrons and positrons in the observable universe, but observations showed electrons comfortably outnumbering positrons by an enormous factor. Wheeler, undaunted, speculated that the "missing" positrons might be hidden inside protons, but Feynman dismissed this as "not a satisfactory explanation."

What Feynman did take seriously — and explicitly credited to Wheeler — was the narrower insight that positrons could be represented as electrons traversing their worldlines backwards in time. This partial extraction of the idea, stripped of its cosmological ambition, became the conceptual nucleus of some of the most important papers in twentieth-century physics. As Feynman said in his **1965 Nobel lecture**:

> *"I did not take the idea that all the electrons were the same one from him as seriously as I took the observation that positrons could simply be represented as electrons going from the future to the past in a back section of their world lines. That, I stole!"*

3. Independent and Parallel Developments: Stueckelberg (1941)

The reinterpretation of antiparticles as time-reversed particles was not solely Wheeler's priority. In **1941**, the Swiss physicist Ernst Carl Gerlach Stueckelberg von Breidenbach independently arrived at the same picture in his paper "La signification du temps propre en mécanique ondulatoire," published in *Helvetica Physica Acta* 14 (1941): 51–80.

Stueckelberg's approach was grounded in classical relativistic mechanics rather than quantum field theory. He took advantage of the general coordinate invariance of the worldline parameter — that is, the freedom to parameterise the worldline by any smooth monotonic function rather than the particle's proper time alone. By choosing a parameterisation in which the worldline could "fold back" in coordinate time, Stueckelberg showed that what a local observer perceives as a positron approaching from the future and annihilating an electron is mathematically identical to a single electron whose worldline, parameterised globally, reverses its time coordinate at the annihilation event.

Stueckelberg's construction was remarkably prescient. He showed explicitly that:

1. A particle's worldline can cross the hypersurface $t = const$ twice, once moving forwards and once backwards, at what an observer would identify as a pair creation or annihilation event.
2. The charge of the time-reversed segment flips sign, yielding positive charge — the positron — from the same equation of motion as the electron, with only the sign of the covariant derivative of proper time changing.
3. This picture can be second-quantised consistently, reproducing the results of hole theory.

The parallel discovery by Stueckelberg was not widely absorbed into the English-language physics community at the time, partly due to World War II and partly due to the language barrier. When Feynman published his 1949 paper, he explicitly acknowledged Stueckelberg's priority, writing that negative energy states "appear in a form which may be pictured (as by Stückelberg) in space-time as waves travelling away from the external potential backwards in time."

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4. From Hypothesis to Formalism: Feynman 1949 and the Theory of Positrons

Feynman's **1949** paper "The Theory of Positrons," published in *Physical Review* 76 (6): 749–759, is where the worldline-reversal picture was given its definitive mathematical form and integrated into quantum theory. The paper's central move is to replace the Dirac "hole theory" — which required a fully occupied negative-energy sea — with a reinterpretation of the **Dirac propagator**.

In hole theory, a positron is the absence of a negative-energy electron. This picture is conceptually awkward, physically unmotivated, and calculationally cumbersome. Feynman showed that the same physical results — pair production, pair annihilation, scattering — follow from choosing boundary conditions on the propagator such that:

- Positive-energy states propagate *forwards* in time (electron moving into the future).
- Negative-energy states propagate *backwards* in time (positron = electron moving into the past).

The **Feynman propagator** $K_+(2,1)$, defined with these boundary conditions, automatically encodes all orders of virtual pair creation and annihilation in a single, compact expression. A single diagram representing "an electron at event 1 scattering to event 2" implicitly contains, depending on the time ordering of events 1 and 2, the interpretations: (a) ordinary forwards scattering; (b) pair annihilation at event 2 if $t_2 < t_1$; and (c) pair production at event 1 if an additional vertex is present.

This is precisely the worldline self-intersection/time-reversal idea made calculational. The propagator $K_+(2,1)$ is the amplitude for a particle to traverse its worldline from spacetime event 1 to spacetime event 2, and the fact that the same amplitude handles both electrons and positrons is the mathematical encoding of Wheeler's original geometric picture, stripped of its cosmological overreach. The **Feynman–Stueckelberg interpretation** of antiparticles remains the standard one in all modern QFT textbooks.

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5. Nambu's Extension and the Standard Interpretation of Antiparticles

Yoichiro Nambu (later a Nobel laureate) took the Wheeler–Feynman–Stueckelberg worldline picture and generalised it to all pair-creation and annihilation events in *Progress of Theoretical Physics* 5 (1950): 614–633. His formulation makes the cosmological aspect of Wheeler's hypothesis into a local, event-by-event statement:

> *"The eventual creation and annihilation of pairs that may occur now and then is no creation nor annihilation, but only a change of directions of moving particles, from past to future, or from future to past."*

Nambu treated all electrons and positrons as segments of worldlines, with **pair creation** corresponding to a worldline turning from backwards-in-time to forward-in-time at a single spacetime event, and pair annihilation corresponding to a worldline turning from forward-in-time to backwards-in-time. In this picture, the total "length" of the worldline is conserved modulo topology, and the number of particles observable at any instant is simply the number of intersections of all worldlines with the corresponding time slice.

Nambu's work cemented the Stueckelberg–Feynman picture as the correct physical framework for understanding antiparticles in relativistic quantum field theory, and its language is the natural one for the worldline formalism discussed in Sections 7–9.

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6. Worldline Self-Intersections: Geometry, Physics, and Interpretation

6.1 What a Worldline Self-Intersection Means

A **worldline self-intersection** occurs when a single particle's spacetime trajectory passes through the same spacetime point x^u more than once — i.e., when $x^u(\tau_1) = x^u(\tau_2)$ for two distinct parameter values $\tau_1 \neq \tau_2$. In Minkowski spacetime, this is a geometrically well-defined event with sharply different physical interpretations depending on context. It is important to distinguish several qualitatively different situations, summarised in Table 1 below.

Table 1. Classification of Worldline Self-Intersection Types

Intersection Type	Description	Physical Interpretation
Timelike self-intersection	Worldline crosses itself at the same spacetime point with both segments timelike.	True self-collision; requires closed timelike curves (CTCs); forbidden in classical GR.
Spacelike close approach	Two segments of the worldline are spatially near simultaneously	Self-Coulomb interaction; treated perturbatively in QED self-energy
Time-reversed crossing	Forward-going and backwards-going segments meet at the same spacetime event.	Pair creation or annihilation event in the Feynman–Stueckelberg picture

Intersection Type	Description	Physical Interpretation
Euclidean self-intersection (instanton)	Worldline in Wick-rotated spacetime forms a closed loop	Schwinger pair production; vacuum decay; topological contribution to effective action
Lightlike coincidence	Two segments of worldline touch on the light cone	Radiation reaction; Liénard-Wiechert field self-interaction

Source: Compiled from Feynman (1949), Stueckelberg (1941), and Schubert (2001).

6.2 Classification of Intersection Types

In the one-electron universe picture, where a single worldline traces all of spacetime, intersections are generic rather than exceptional. The worldline $x^u(\tau)$ is parameterized by an arbitrary monotonic parameter τ (which Stueckelberg first systematically exploited), and the physical time coordinate $x^0(\tau)$ is allowed to be non-monotonic. Every time $x^0(\tau)$ achieves a local maximum or minimum — i.e., every time $\dot{x}^0(\tau) = 0$ and changes sign — the worldline has "turned around in time," generating what observers perceive as a pair creation or annihilation event.

For a worldline of total parameter length T with N time-reversals, a fixed-time slice at generic t sees approximately $N/2 + 1$ intersections, half identified as electrons and half as positrons. The net winding number $w = (electrons) - (positrons)$ is conserved globally and equals the total charge in units of e. This is the rigorous formalization of Wheeler's time-slice counting argument.

6.3 Observable Consequences of Intersections

The immediate physical consequence of a time-reversal intersection is a pair creation or annihilation event. In standard QED: a worldline turning from future-directed to past-directed at event x corresponds to **pair annihilation** — an electron and positron converge at x and annihilate, typically producing two photons (to conserve energy-momentum and angular momentum). A worldline turning from past-directed to future-directed at event x

corresponds to **pair production** — a sufficiently energetic photon (or electromagnetic field) induces an electron-positron pair to materialize at x.

The observable signatures of these intersection events are among the most well-measured phenomena in particle physics: the **511 keV** gamma-ray lines from positron-electron annihilation (the basis of PET imaging), Bhabha scattering cross-sections (measured to better than **10^{-4}** precision at LEP), and pair production rates from high-energy photons. Every such measurement is indirect confirmation of the physical content of worldline time-reversals — the structural element that Wheeler's hypothesis required. What the one-electron hypothesis adds beyond standard QED is the claim that all these intersection events belong to the same worldline. This claim has no observable consequence distinguishable from standard QED at the level of individual scattering events; its only testable prediction is global (the matter-antimatter equality constraint), which it fails.

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7. Worldline Self-Intersections in Quantum Field Theory

7.1 The First-Quantized (Worldline) Path Integral

The modern worldline formalism treats quantum field theory amplitudes as quantum mechanics on the particle's worldline rather than as field operations in a Hilbert space. The bridge between the two descriptions is the proper-time representation of the propagator:

$$\langle x_2 |\ 1/(-\partial^2 + m^2)\ | x_1 \rangle = \int_0^\infty dT\ \int[x(0)=x_1 \text{ to } x(T)=x_2]\ Dx^\mu(\tau)\ \exp\left(-\int_0^T [\dot{x}^2/4 + m^2]\ d\tau\right)$$

This is a path integral over all worldlines connecting x_1 to x_2 in proper time T, weighted by the relativistic action. The **effective action** (generating functional for one-loop corrections) is obtained by taking the trace over closed worldlines — precisely the loops of the Feynman diagram expansion.

7.2 Schwinger Proper Time, the Heat Kernel, and Fock's Representation

The proper-time representation has a distinguished history. Fock first introduced a proper-time parameterisation of the electron propagator in **1937**; Schwinger systematised it in **1951** as the "heat kernel" representation of the propagator, obtaining the celebrated **Euler-Heisenberg effective Lagrangian** for QED in a constant electromagnetic field:

$$\mathcal{L}_{EH} = (1/8\pi^2) \int_0^\infty (dT/T^3)\, e^{-m^2T} [eET \coth(eET) - 1 - (eET)^2/3]$$

The imaginary part of this Lagrangian encodes the rate of pair production (Schwinger mechanism), nonperturbative in the electric field E:

$$P \sim \exp(-\pi\, m^2/(eE))$$

The exponential suppression means pair production is negligible for E much less than $E_{Schwinger} = m^2/e \approx \mathbf{1.3 \times 10^{18}\ V/m}$ — a field strength that has not yet been achieved experimentally. The worldline path integral provides a particularly transparent derivation of this result: the imaginary part arises from closed worldline loops that are real saddle points of the Euclidean action — the worldline instantons.

7.3 Worldline Instantons and Pair Production

Worldline instantons are solutions to the classical equations of motion in Euclidean spacetime ($t \to -i\tau_E$) that are closed loops. For a constant electric field E in the z-direction, the instanton is a circle in the (τ_E, z) plane with radius $r = m/(eE)$. This closed loop is precisely a worldline that: (1) begins as a positron moving backwards in coordinate time; (2) reaches a turning point and becomes an electron moving forwards in coordinate time; and (3) returns to its starting point — the closed loop represents a virtual pair that is created and annihilated by the background field.

The saddle-point approximation to the worldline path integral gives the leading exponential $\exp(-\pi\ m^2/(eE))$, whilst higher-order corrections (preexponential factors) require the ratio of functional determinants around the instanton. Recent work has generalised this framework to inhomogeneous, multi-pulse electromagnetic fields using **discrete worldline instantons**—numerically computed saddle points in multidimensional fields. For fields with N stationary points, there are N instantons at the amplitude level, yielding $N(N+1)/2$ contributions at the probability level due to interference terms.

In recent work (Degli Esposti & Torgrimsson, *Phys. Rev. D* 112, 016026, **2025**), worldline instantons in spatiotemporal fields have been shown to produce **Moiré interference patterns** in the electron-positron momentum spectrum and **Aharonov-Bohm phases** depending on the spatial separation of field pulses. These represent direct, measurable consequences of the topology of worldline self-approaches in background fields.

7.4 Do Worldline Self-Intersections Correspond to Observable Collisions?

This question requires careful disambiguation. In the worldline path integral, each path $x^u(\tau)$ is a single configuration being integrated over. A self-intersection within one such path is interpreted differently depending on the context of the calculation. In *Euclidean spacetime*, a topologically distinct class of paths contributes to the path integral with a different weight. Worldlines that cross themselves form a set of measure zero in the path-integral measure (for smooth paths), but they carry topological information encoded in knot invariants. In the worldline instanton approach, the instanton loop itself is a self-intersection in the sense that its start and end points coincide— that is, a closed loop. This does not correspond to a physical particle-particle collision; it corresponds to a quantum tunnelling event (pair production).

In *Minkowski spacetime* (real time), a genuine self-intersection requires a single particle to be at the same spatial location at two different times, possible only via a non-monotonic worldline (time reversal). This is the pair-creation/annihilation event by Stueckelberg and Feynman. In the *one-electron universe picture*, every pair production event is reinterpreted as the single worldline turning a corner in time — the electron "collides with itself" in the sense that the same worldline passes through the same neighbourhood of spacetime at two distinct proper times. The observable outcome — two **511 keV** photons emitted back-to-back in pair annihilation — is entirely indistinguishable from standard QED. No additional observable effect arises from worldline self-intersections beyond what the Feynman diagram already predicts.

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8. The Bern–Kosower–Strassler Worldline Formalism

8.1 Origins in String Theory

The worldline formalism was dramatically reinvigorated in the late 1980s and early 1990s by **Zvi Bern** and **David Kosower**, who worked on the infinite-string-tension limit of string theory amplitudes. Their insight was that string theory, taken to the limit where the string tension becomes infinite (all massive string modes decouple), reduces to ordinary QFT, but in an integral representation far more compact than the Feynman diagram expansion. Bern and Kosower derived **master formulae** for N-gluon one-loop amplitudes in Yang-Mills theory — parameter-integral representations that encode all helicity and colour information in a single expression, without reference to individual Feynman diagrams. These Bern-Kosower rules are equivalent to the sum of all relevant Feynman diagrams but require fewer terms dramatically: the 4-gluon amplitude, for example, involves **4 Feynman diagrams** in standard methods but is captured by a **single** Bern-Kosower integral.

8.2 Master Formulae and One-Loop Amplitudes

Strassler (1992) showed that the Bern-Kosower rules have a direct first-quantised derivation from the worldline path integral, without reference to string theory at all. The one-loop N-photon amplitude in spinor QED can be written as a path integral average over closed worldlines of proper-time length T of products of photon vertex operators $V_\mu(k, \tau) = \varepsilon_\mu \dot{x}^\mu(\tau)$

exp(ik·x(τ)). Table 2 summarises the principal advantages of the worldline approach over conventional Feynman methods.

Table 2. Comparison: Conventional Feynman Diagrams vs Worldline Formalism

Feature	Conventional Feynman Diagrams	Worldline Formalism
Photon symmetry	Must sum over crossed diagrams separately	Manifest Bose symmetry of all external photons
Dirac algebra	Extensive gamma-matrix traces required	Replaced entirely by Grassmann integrals
Spin effects	Separate spinor and scalar QED calculations	Unified via worldline SUSY; single framework
Multi-photon scaling	Diagram number grows factorially with *N*	Single master formula valid for all *N*
Gauge invariance	Verified diagram by diagram	Manifest via integration by parts on the loop

Source: Strassler, M. J. (1992); Schubert, C. (2001).

8.3 Multi-Loop Extensions and Graph-Based Green's Functions

Beyond one loop, the worldline formalism requires **multi-loop worldline Green's functions**, which are constructed from the graph-theoretic structure of the loop diagram (the Schwinger-Dyson skeleton). Schubert and collaborators developed this program systematically through the 1990s and 2000s. The two-loop quenched QED photon propagator was reproduced from worldline methods by Bern and Dunbar, demonstrating that the formalism is not merely a rearrangement at one loop but a genuinely new calculational framework. Recent advances (arXiv:2503.08846, **2025**) extend the formalism to extended worldline supersymmetry with *N* supercharges, where the worldline gauge multiplet contains an einbein *e(τ)* (gauge field for worldline reparametrizations), gravitinos $\chi^I(\tau)$ (gauge fields for the *N* worldline supersymmetries), and a gauge field *a(τ)* for the R-symmetry rotating worldline fermions. This extended structure captures particles of arbitrary spin in a unified first-quantised framework.

9. Grassmann Variables and the Spinning Worldline

9.1 Fradkin's Representation of the Spin Factor

The essential challenge in applying the worldline path integral to spin-1/2 particles is encoding the spin degrees of freedom. For a spinless particle, the path integral is purely bosonic ($x^\mu(\tau)$). For a spin-1/2 particle, the Dirac equation demands a 4-component spinor structure. **Fradkin (1966)** showed that the spin factor — a path-ordered exponential of the field strength tensor contracted with the Lorentz generators — can be represented as a **Grassmann path integral** over anticommuting, Lorentz-vector-valued functions $\psi^\mu(\tau)$ satisfying antiperiodic boundary conditions $\psi^\mu(T) = -\psi^\mu(0)$. The Grassmann fields $\psi^\mu(\tau)$ carry the spin degrees of freedom geometrically: their anticommutativity encodes the Pauli exclusion principle at the level of the path integrand, whilst the coupling $F_{\mu\nu}\psi^\mu\psi^\nu$ generates the spin-magnetic moment interaction. The antiperiodic boundary condition selects the fermionic sector (R sector in string language) over the bosonic sector (NS sector).

9.2 The N=1 Worldline Supersymmetry Algebra

The combined bosonic-Grassmann action possesses a **worldline supersymmetry** with a single supercharge Q = ψμ ẋμ, which squares to the Hamiltonian $Q^2 = H = -\dot{x}^2/2 + m^2/2$. The worldline action is invariant under the SUSY transformation:

$$\delta x^\mu = \varepsilon\, \psi^\mu, \quad \delta\psi^\mu = -\varepsilon\, \dot{x}^\mu$$

for anticommuting parameter ε. This **N=1 worldline supersymmetry** is the first-quantised analogue of spacetime supersymmetry; it ensures the correct Dirac equation for the quantised particle and generates the Pauli exclusion principle via the Grassmann statistics of ψ^μ. Extending to **N=2** (two Grassmann directions $\psi^\mu, \bar{\psi}^\mu$) yields the bosonic spinning particle — a spin-1 system, relevant for Maxwell field propagation. **N=4** yields spin-2 (graviton propagation). This hierarchy of worldline supersymmetry is the first-quantised analogue of the supersymmetric multiplet structure and provides a unified framework for particles of all spins, without introducing spacetime supersymmetry.

9.3 Colour Grassmann Variables and Non-Abelian Generalisation

For non-Abelian gauge theories (QCD, Yang-Mills), the worldline must carry colour degrees of freedom in addition to spin. Bern, Kosower, and subsequently Strassler showed that colour can be incorporated by adding a set of **Grassmann variables** $c^a(\tau), \bar{c}^a(\tau)$ in the appropriate representation of the gauge group. Integration over the colour Grassmann variables yields Wilson lines and colour traces automatically, reproducing the full non-Abelian amplitude structure. In the context of the one-electron universe, the colour Grassmann variables provide an intriguing extension: if the single worldline carries colour as well as electric charge, the strong-force dynamics of quarks and gluons could, in principle, be described as topological structures of the same worldline. This remains speculative, but the mathematical framework exists.

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10. Topological Approaches: Knots, Braids, and Invariants

10.1 The Worldline as a Knot in Spacetime

The one-electron universe picture implicitly requires the single worldline to be a **knot** — a closed or infinite curve in four-dimensional spacetime whose topology encodes physically observable quantities. The language of knot theory, developed rigorously in the 20th century, provides the natural mathematical framework for this analysis. A worldline that closes on itself — either as a particle loop in a Feynman diagram or as a worldline instanton — is mathematically a closed curve in 3+1 dimensions. In four spacetime dimensions, generically positioned curves do not intersect (they have "room" to miss each other), but worldline crossings and self-linkings can still be characterised topologically by projection onto 3D slices. The relevant topological invariants are summarised in Table 3.

Table 3. Topological Invariants of Worldlines and Their Physical Interpretations

Invariant	Definition	Physical Interpretation
Gauss linking number $Lk(L_1, L_2)$	Double line integral over the Gauss formula	Topological EM phase; Aharonov-Bohm analogue
Self-linking number (writhe)	Linking number of a curve with a parallel copy	Worldline framing; related to anomalous magnetic moment
Jones polynomial $V_L(t)$	Skein-relation polynomial from Reidemeister moves	Non-Abelian statistics; quantum group structure
Alexander polynomial	Determinant of Seifert matrix	Genus of Seifert surface; charge conservation topology
Winding number	Net crossings with time hyperplane	Total conserved charge (net electrons minus positrons)

Source: Jones (1985); Atiyah (1988); Bizri (2025).

10.2 Gauss Linking Number and the Anomalous Magnetic Moment

The most striking recent development in topological worldline physics is the identification of the **anomalous magnetic moment** — the Schwinger correction $(g-2)/2 = \alpha/(2\pi) \approx 1.16 \times 10^{-3}$ — with a topological self-linking contribution from the worldline. In the monadic electron framework (Bizri, 2025), the one-loop QED correction to the electron's magnetic moment arises when the worldline curves back and interacts with itself, forming a loop configuration characterised by the **self-linking number (writhe)** of the worldline. The result:

$$\Delta\mu = (\alpha/2\pi)\, \mu_B = (e^2/8\pi^2 m) \cdot Wr(\text{worldline loop})$$

matches Schwinger's famous 1948 result exactly, but derives it from the topology of the worldline geometry rather than from perturbative loop integrals. This is a striking demonstration of how purely geometric, topological reasoning can reproduce a quantitative result previously obtained only through lengthy Feynman diagram calculations.

10.3 Jones Polynomial and Topological Quantum Numbers

The **Jones polynomial** $V_L(t)$, discovered by Vaughan Jones in 1984, is a knot invariant defined via the skein relation and computable from the R-matrix of a quantum group. In the TQFT context, it appears as the expectation value of a Wilson loop operator in Chern-Simons gauge theory. In the worldline context, the Jones polynomial encodes how the worldline self-entangles in a gauge-theory-dependent way. For the one-electron universe hypothesis, if the single worldline is a particular knot L, its Jones polynomial $V_L(t)$ is a topological invariant of the universe that would, in principle, be measurable through subtle correlations in quantum-interference experiments. Bizri (2025) proposes that distinct knot types of the worldline correspond to distinct topological quantum numbers — new conserved quantities beyond those of the Standard Model — and that these might predict small but detectable energy shifts in atomic spectroscopy.

10.4 Braid Group, Statistics, and the Pauli Exclusion Principle

In 2+1 dimensions, particle worldlines are curves in 3-dimensional spacetime, and their mutual braiding defines the **braid group** B_n for n particles. The representation theory of B_n determines the statistical phase acquired when two particles are exchanged: trivial phase (0) for bosons, phase π for fermions, and arbitrary phase for anyons. In 3+1 dimensions, worldlines cannot be knotted around each other in a topologically stable way (because there is an extra dimension for curves to avoid each other), and the statistics group reduces to the symmetric group S_n, reproducing ordinary Bose-Einstein or Fermi-Dirac statistics.

The **Pauli exclusion principle** for electrons — no two electrons in the same quantum state — is in the worldline picture equivalent to the topological impossibility of two worldline segments occupying the same spacetime trajectory simultaneously when both carry anticommuting Grassmann spin structure. For the one-electron universe, where there is only one worldline, the Pauli exclusion principle becomes the statement that the worldline cannot self-intersect in 3D space in a topologically non-trivial way — a constraint enforced by the braid group relations and the **Yang-Baxter equation**. The fermionic statistics of the electron thus emerge as a geometric constraint on the worldline's topology rather than a separately postulated quantum principle.

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11. Modern Reinterpretations of the One-Electron Universe

11.1 The Monadic Electron Framework (Bizri, 2025)

The most technically developed recent revival of the one-electron universe idea is the **Monadic Electron Universe** framework proposed by Sam Bizri in two papers (April 2025). The framework starts from the standard worldline action for a spinning particle interacting with an electromagnetic field, augmented with Grassmann spin variables and topological terms—specifically, delta-function kink insertions at proper times when the worldline undergoes a time reversal. These **kink terms** mark pair creation/annihilation events and contribute boundary terms to the path integral that flip the sign of the charge.

The framework reproduces all one-loop QED results and interprets the vacuum as a Planck-scale foam of virtual worldline loops. From worldline topology alone it derives: Dirac spinors (from the N=1 worldline supersymmetry generated by ψ^μ); Fermi-Dirac statistics (from the anticommuting nature of Grassmann fields); the anomalous magnetic moment (from the self-linking number of the one-loop worldline configuration); the Pauli exclusion principle (from the geometric impossibility of worldline self-intersection with coherent spin structure); and pair production (from worldline kinks/time-reversals). Bizri's second paper extends the framework to non-Abelian gauge theories by adding colour Grassmann variables, and to gravity by treating the worldline curvature as generating a long-range strain in the "braid network" of spacetime.

Predicted experimental signatures include: spectroscopic shifts of order $\mathbf{\Delta E/E \sim 10^{-19}}$ in atomic systems (near the current sensitivity frontier of optical clocks); phase shifts $\mathbf{\Delta\phi \sim \lambda_C / L}$ in electron interferometers with path length $L \sim 1$ cm (approximately $\mathbf{10^{-12}}$ **rad** — at the edge of current sensitivity); and corrections to $g-2$ at order $\mathbf{\Delta(g-2)/(g-2) \sim 10^{-14}}$, within the target precision of proposed next-generation measurements.

11.2 The Electron Monad and Quantum Gravity

The holographic and quantum-gravity implications of the monadic picture are speculative but structurally interesting. If the electron's worldline is the fundamental object of the theory, then spacetime itself might be emergent from the worldline's braiding structure — an idea with resonances in holography (AdS/CFT), loop quantum gravity, and quantum graphity. The worldline entropy in this picture scales with the horizon area, satisfying the Bekenstein-Hawking relation $S = A/(4\ell_P{}^2)$, which is consistent with, but not derived from, the holographic principle. These speculative extensions are not experimentally distinguished from the Standard Model at present, but they point towards a conceptually unified picture of matter and spacetime.

11.3 Biquaternionic Analysis and the Implicit Unique Worldline

A mathematically distinct approach represents the electron worldline implicitly through **biquaternion-valued functions** $F(Z)$ in the algebra B of biquaternions (complexified quaternions). The B-differentiability condition generalises the Cauchy-Riemann equations to the non-commutative setting. The resulting equations are Lorentz-invariant on the Minkowski subspace of C^4, possess natural spinor and gauge structures, and reduce to a nonlinear generalisation of the Cauchy-Riemann equations. The implicit worldline in this framework is defined as the **singular locus** of the biquaternionic function — the set of points where the Jacobian determinant vanishes. This singular locus is the caustic set of the worldline, where multiple worldline sheets converge, corresponding physically to pair-creation/annihilation events. The framework provides a new global, coordinate-free description of the one-electron worldline that is fully consistent with relativistic covariance.

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12. Critical Analysis: Strengths, Weaknesses, and Open Problems

12.1 The Matter–Antimatter Asymmetry Objection

The central, and still decisive, empirical objection to the one-electron universe hypothesis is the observed **matter-antimatter asymmetry**. Every time-symmetric single worldline that begins and ends at the temporal boundaries of the universe must have equal numbers of forward-moving (electron) and backwards-moving (positron) segments at any given cosmic time. This requires $n_{electrons} = n_{positrons}$ throughout cosmic history, in sharp contradiction with observation.

Quantitatively, the baryon-to-photon ratio is measured to be $\boldsymbol{\eta = n_b/n_\gamma \approx 6.1 \times 10^{-10}}$ (from BBN and CMB), and the lepton sector shows a comparable asymmetry (about **10^{10} electrons per positron** in the observable universe). Wheeler's suggestion that the missing positrons were hidden inside protons is incompatible with the known quark structure of the proton established by deep inelastic scattering experiments at SLAC and HERA. Modern cosmological explanations for matter-antimatter asymmetry — baryogenesis via electroweak sphaleron processes, leptogenesis from heavy Majorana neutrino decay, or the Affleck-Dine mechanism — all require specific CP-violating physics that is orthogonal to and incompatible with the one-electron universe in its literal form. The monadic electron reinterpretations acknowledge this objection by attributing the asymmetry to the "initial conditions of the worldline topology," but this is a placeholder assertion rather than a derived result. Table 4 summarises the observational incompatibilities.

Table 4. Observational Compatibility of the One-Electron Hypothesis

Quantity	Observed Value	One-Electron Prediction	Compatible?
$n(e^-)$ / $n(e^+)$	~10^{10}	= 1 (exactly)	**No**
Baryon-to-photon ratio	6.1×10^{-10}	0 (no net baryons)	**No**
Proton internal structure	3 quarks (uud)	Electron-positron composite	**No**
Leptogenesis requirement	CP violation needed	Single worldline = CP symmetric	**No**

Source: PDG (2024); BBN constraints; deep inelastic scattering data (SLAC, HERA).

12.2 Compatibility with the Standard Model

Modern reinterpretations of the one-electron universe within the worldline formalism are generally *consistent with* but do not *derive* the Standard Model. They reproduce QED exactly (guaranteed by the equivalence of the worldline formalism to second-quantised QED proven by Strassler in 1992). For QCD, the colour Grassmann variable extension is consistent with asymptotic freedom and confinement at a formal level. However, the actual calculation of the QCD spectrum (hadronic masses, confinement radius) requires the full non-perturbative machinery of lattice QCD or AdS/CFT. The weak force and Higgs mechanism present additional challenges, as the worldline formalism is most natural for gauge bosons and fermions in external fields, not for spontaneous symmetry breaking.

12.3 Observational and Experimental Status

The experimental status of the one-electron universe hypothesis *as a hypothesis* (rather than as the broader worldline formalism) is negative: no observation distinguishes it from ordinary QED, and its most specific prediction (matter-antimatter equality) is falsified. The worldline formalism, by contrast, is in excellent experimental health — it has been used to compute QED and QCD amplitudes in agreement with experiment to many decimal places. The topological extensions (Sections 10–11) make novel, specific predictions that are, as of 2026, untested but not yet falsified. Table 5 provides a quantitative overview of experimental sensitivities.

Table 5. Experimental Sensitivity to Topological Worldline Corrections

Observable	Standard QED Prediction	Topological Correction (est.)	Current Sensitivity	Near-Future Reach
Electron g–2	1.00115965218...	$\delta \sim 10^{-14}$	$\delta \sim 10^{-13}$	10^{-14} (proposed)
H atom ground-state energy	−13.6056... eV	$\Delta E/E \sim 10^{-19}$	10^{-18} (optical clocks)	10^{-19} (next-gen)
Electron interferometer phase	0 (std QED)	$\Delta\phi \sim 10^{-12}$ rad	$\sim 10^{-11}$ rad	10^{-12} rad (proposed)
Vacuum birefringence	PVLAS: $< 10^{-21}$	Topological $\sim 10^{-22}$	10^{-21} (PVLAS)	10^{-22} (ATLAS)

Source: Bizri (2025); Degli Esposti & Torgrimsson (2025); Gabrielse group (2023).

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13. Outlook and Key Takeaways

Short-Term Outlook (2026–2030)

The primary experimental battleground for topological worldline predictions is the **electron anomalous magnetic moment**. The current experimental value from Gabrielse's Harvard group (2023) agrees with the Standard Model prediction to **13 significant figures**. Proposed next-generation experiments targeting the 14th digit would either constrain or begin to probe the topological corrections predicted by the monadic electron framework. Simultaneously, next-generation optical atomic clocks operating at the **10^{-19} frequency level** could probe the spectroscopic shifts predicted by worldline topology.

Medium-Term Outlook (2030–2040)

The worldline instanton program for Schwinger pair production will receive its first direct experimental test if laser facilities (ELI-NP, LUXE at DESY, or successors) reach field strengths approaching **10^{18} V/m**. The Moiré interference patterns and Aharonov-Bohm phases predicted by Degli Esposti and Torgrimsson (2025) in the electron-positron momentum spectrum provide a new class of observable signatures of worldline topology in strong-field physics — directly testing the instanton formalism in ways that probe the geometry of worldline self-approaches.

Long-Term Outlook (2040+)

The question of whether a single-worldline description of matter is a reinterpretation of standard physics (empirically equivalent, computationally superior in some regimes) or a theory with new predictions (distinguishable from the Standard Model) hinges on whether topological quantum numbers beyond the Standard Model's $U(1) \times SU(2) \times SU(3)$ charges can be identified and measured.

Numbered Actionable Takeaways

4. **Treat the worldline formalism as a calculational tool, not just a philosophical reinterpretation.** For multi-photon QED amplitudes and one-loop effective actions in background fields, it is strictly superior to conventional Feynman diagram methods — use it for any calculation involving more than 4 external photons at one loop.
5. **Follow the worldline instanton literature for strong-field physics.** The Degli Esposti-Torgrimsson 2025 results on Moiré patterns and Aharonov-Bohm phases in Schwinger pair production represent the most direct connection between worldline topology and near-future experiment.
6. **Track next-generation g–2 proposals.** Any improvement beyond the 10^{-13} level will begin testing the topological corrections predicted by the monadic electron model — if such corrections are real, they are the most accessible experimental signature of the one-electron universe picture.
7. **Engage with the biquaternionic worldline program.** The implicit worldline formulation (Section 11.3) offers a potentially new global description of worldline

topology that could unify the caustic structure of worldlines with the pair-production physics — mathematically rich and underexplored.

8. **Revisit the matter-antimatter asymmetry constraint** before advancing any literal version of the one-electron universe hypothesis. Any serious revival must provide a mechanism for the worldline topology to generate $n(e^-)/n(e^+) \approx 10^{10}$ from first principles, without which the hypothesis remains observationally ruled out.
9. **Explore Grassmann colour variable extensions for QCD worldline calculations.** The non-Abelian generalisation is technically well-defined and has been used for one-loop amplitudes; higher-loop extensions with colour Grassmann variables could yield new insights into quark confinement and asymptotic freedom from a first-quantised perspective.
10. **Connect worldline topology to quantum information.** The braid group structure of worldlines (Section 10.4) is formally identical to the anyon statistics used in topological quantum computation. The Kitaev-Preskill program for fault-tolerant quantum computing using non-Abelian anyons is, in a precise mathematical sense, a direct application of worldline braiding physics — a connection that deserves more attention in the foundational physics literature.

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